

A PRIMITIVE WEIRD NUMBER WITH TWO SQUARE ODD PRIME FACTORS AND $\Omega = 8$

ABSTRACT. A *weird* number is an abundant number that is not semiperfect, and a *primitive weird number* (PWN) is a weird number no proper divisor of which is weird. For $n \in \mathbb{N}$ let Ω_n denote the least value of $\Omega(m)$, the number of prime factors of m counted with multiplicity, over all PWN m having exactly n square odd prime factors. Open Question 1 of Amato, Hasler, Melfi and Parton, quoted here from arXiv:1802.07178v2, records $\Omega_1 = 7$, $8 \leq \Omega_2 \leq 12$, $8 \leq \Omega_3 \leq 15$, and $\Omega_n \geq 8$ for $n \geq 2$. We exhibit

$$m = 2^2 \cdot 13^2 \cdot 19^2 \cdot 46219 \cdot 1108619 = 12504224434300196,$$

a PWN with exactly two square odd prime factors and $\Omega(m) = 8$, so $\Omega_2 \leq 8$. With their lower bound $\Omega_2 \geq 8$, which is quoted and not reproved here, this gives $\Omega_2 = 8$ and shows that their bound $\Omega_n \geq 8$ for $n \geq 2$ is attained; the equality is conditional on that quoted bound. The verification is exact integer arithmetic: two primality checks by trial division, then $\sigma(m) - 2m = 8$ with the divisors of m that are at most 8 being 1, 2, 4, whose total is 7, and five comparisons showing the maximal divisors of m deficient.

1. THE QUESTION

Throughout, σ is the sum-of-divisors function, $\Omega(m)$ is “the number of prime factors counted with their multiplicity” and $\omega(m)$ is the number of distinct prime factors. Write

$$\Delta(m) = \sigma(m) - 2m,$$

so that m is *abundant* exactly when $\Delta(m) > 0$ and *deficient* exactly when $\Delta(m) < 0$. Following Amato, Hasler, Melfi and Parton [1], a number is *semiperfect* if it is expressible as a sum of distinct proper divisors of itself; a *weird* number is “a number which is abundant but not semiperfect”; and a *primitive weird number* (PWN) is “a weird number which is not a multiple of any smaller weird number”. A *square odd prime factor* of m is an odd prime p with $p^2 \mid m$.

In the section “Open problems” of [1] the following appears, as the first numbered Open Question of that section. It and the theorem below are quoted from arXiv:1802.07178v2, which was not compared with the published version.

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Open Question 1 ([1]). *For each n in $\mathbb{N}$, find a PWN with exactly n square odd prime factors and the least $\Omega = \Omega_n$. From the previous section and [the theorem quoted below] we obtain $\Omega_1 = 7$, $8 \leq \Omega_2 \leq 12$, $8 \leq \Omega_3 \leq 15$, and in general if $n \geq 2$ we have $\Omega_n \geq 8$.*

The lower bounds quoted there are the authors' own theorem, which reads verbatim and in full: “There are no PWN m with a quadratic or higher power odd prime factor and $\Omega(m) < 7$. There are no PWN m with 2 quadratic odd prime factor and $\Omega(m) = 7$. There are no PWN m with a cubic or higher power odd prime factor and $\Omega(m) = 7$.” The bound $\Omega_2 \geq 8$ is taken here from that theorem as a whole, transcribed and not reproved. We do not apportion it among the three sentences: they distinguish a quadratic from a cubic or higher odd prime factor, whereas the definition of square odd prime factor used here, $p^2 \mid m$, does not, and we do not settle which reading of “quadratic odd prime factor” the source intends. The upper bound $\Omega_2 \leq 12$ is the smallest entry of the Ω column of their table of PWN with more than one square odd prime factor, which reads 12, 12, 12, 13, 14, 15.

2. THE WITNESS

Theorem 1. *Let*

$$m = 2^2 \cdot 13^2 \cdot 19^2 \cdot 46219 \cdot 1108619 = 12504224434300196.$$

Then m is a primitive weird number, $\Omega(m) = 8$, and m has exactly two square odd prime factors, namely 13 and 19. Consequently $\Omega_2 \leq 8$, and with the lower bound $\Omega_2 \geq 8$ quoted from [1] in §1, $\Omega_2 = 8$.

The following criterion is a proposition of [1], where it reads: “An abundant number n is weird iff $\Delta(n)$ cannot be expressed as a sum of distinct proper divisors of n ”. A proof is included for completeness.

Lemma 2. *An abundant number N is weird if and only if $\Delta(N)$ is not a sum of distinct proper divisors of N .*

Proof. Let P be the set of proper divisors of N , so $\sum_{d \in P} d = \sigma(N) - N = N + \Delta(N)$. If $S \subseteq P$ then its complement $T = P \setminus S$ satisfies $\sum_{d \in T} d = N + \Delta(N) - \sum_{d \in S} d$. Hence $\sum_{d \in S} d = N$ holds for some $S \subseteq P$ if and only if $\sum_{d \in T} d = \Delta(N)$ holds for some $T \subseteq P$. The left-hand condition is semiperfection and the right-hand one is representability of $\Delta(N)$; an abundant number is weird exactly when the former fails. $\square$

Proof of Theorem 1. (i) The factorisation. 46219 and 1108619 are prime: neither is divisible by any prime up to $\lfloor \sqrt{46219} \rfloor = 214$, respectively $\lfloor \sqrt{1108619} \rfloor = 1052$. Hence $\Omega(m) = 2 + 2 + 2 + 1 + 1 = 8$ and $\omega(m) = 5$, and the odd primes occurring in m to an exponent at least 2 are exactly 13 and 19: two of them, both to the exponent exactly 2, so the two readings of “square odd prime factor” discussed in §1 agree for m .

(ii) *Abundance.* By multiplicativity,

$$\sigma(m) = 7 \cdot 183 \cdot 381 \cdot 46220 \cdot 1108620 = 25008448868600400,$$

$$2m = 25008448868600392,$$

so

$$\Delta(m) = 8 > 0.$$

(iii) *Weird.* The smallest odd prime factor of m is $13 > 8$, so the divisors of m that are at most $\Delta(m) = 8$ are exactly 1, 2 and 4, and

$$1 + 2 + 4 = 7 < 8.$$

No sum of distinct divisors of m can therefore equal 8 — the eligible divisors do not even total 8 — so by Lemma 2 the abundant number m is weird.

(iv) *Primitive abundant.* The abundancy $\sigma(x)/x$ is non-decreasing along divisibility, so every proper divisor of m is deficient as soon as the five maximal ones are, and these are settled by exact integer comparison:

x	$\sigma(x)$	$2x$
$m/2$	10717906657971600	12504224434300196
$m/13$	1913214667543200	1923726836046184
$m/19$	1312779468168000	1316234150978968
$m/46219$	541074185820	541085892568
$m/1108619$	22558179420	22558199768

Each left entry is smaller than the right entry beside it, so each m/q is deficient and m is primitive abundant. In particular no proper divisor of m is abundant, hence none is weird, and m is a *primitive* weird number.

(v) *The closure.* m is a PWN with exactly two square odd prime factors and $\Omega(m) = 8$, so $\Omega_2 \leq 8$; the theorem of [1] quoted in §1 gives $\Omega_2 \geq 8$; hence $\Omega_2 = 8$. $\square$

Corollary 3. *Granting the bound $\Omega_n \geq 8$ recorded for $n \geq 2$ in Open Question 1 of [1], it is attained at $n = 2$.*

Status. Only the case $n = 2$ of Open Question 1 is settled. The bound $\Omega_2 \leq 8$ is proved above; the matching $\Omega_2 \geq 8$ is the theorem of [1] quoted in §1, transcribed and not reproved, so the equality $\Omega_2 = 8$ is conditional on that theorem. Nothing here bounds Ω_n for $n \geq 3$.

The integer itself is not new: 12504224434300196 appears as the tenth term of OEIS A063788 [2], “numbers k such that $\sigma(k) = 2k + \Omega(k)$ ”, consistently with $\Delta(m) = 8 = \Omega(m)$ above. We make no priority claim for the observation that m is a PWN with two square odd prime factors, nor for the closure that follows from it; we have not established that either is unrecorded.

REFERENCES

- [1] G. Amato, M. F. Hasler, G. Melfi, and M. Parton, *Primitive abundant and weird numbers with many prime factors*, J. Number Theory **201** (2019), 436–459. doi:10.1016/j.jnt.2019.02.027; arXiv:1802.07178v2.
- [2] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*: sequence **A063788**, numbers k with $\sigma(k) = 2k + \Omega(k)$, oeis.org/A063788.